

YASHWANTH IRATLA

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Education

University of California, Berkeley

B S Mechanical Engineering

Berkeley, CA

Expected May 2029

Experience

Liberty Robotics — FRC Team 4131

President

Renton, WA

Apr 2022 – May 2026

Lead a 40-member FIRST Robotics Competition team across robot development, mechanical design, manufacturing, outreach, and technical training.

Designed a deployable multi-level climbing mechanism for the 2026 robot within a 115-lb weight limit and 30-in. extension constraint.

Managed annual summer camp operations and outreach programs generating approximately 50K annually for team funding.

Penn State Research

Research Intern

Remote

Jun 2024 – Present

Developed an iOS application using Swift and Apple HealthKit for wearable health data collection and analysis.

Applied CNN, gradient boosting, and SVM models to wearable health telemetry and contributed to a technical research paper.

Verveba Telecom

Engineering Intern

Dallas, TX

Aug 2024

Created test cases for a network organization tool used by U.S. telecommunications carriers.

Analyzed telecom systems and assisted with patent development for a network optimization product in a Scrum/Jira workflow.

Engineering Projects

Three-Stage Cascading Elevator *CAD Mechanical Design Manufacturing*

Designed and assembled a three-stage cascading elevator with fully belt-driven rigging, combining vertical extension and arm pivot to reach a 7-ft-high algae scoring target.

Optimized belt tensioning, pulley support, and stage synchronization through iterative prototyping, custom 3D-printed components, and aluminum fabrication.

Deployable Multi-Level Climbing Mechanism *CAD Mechanical Design FEA*

Designed a deployable climbing mechanism using a dead-axle pivot, belt/gear transmission, chain drive, and carbon-ber hooks to climb multiple bars.

Balanced 115-lb weight, 12-in. extension, packaging, and structural constraints through FEA, prototyping, and extensive hook testing.

MK4 Swerve Drivebase & Robot Architecture *CAD Packaging Manufacturing*

Designed frame rails, cross members, bumper mounts, battery mount, and bellypan architecture around an MK4 swerve drivetrain.

Integrated an under-bumper intake and relocated electronics beneath the bellypan to improve subsystem packaging, wiring, and serviceability.

5" FPV Drone *Mechanical Assembly Electrical Integration Troubleshooting*

Designed and assembled a 5-inch FPV drone by integrating the frame, motors, flight controller, ESC, ELRS receiver, battery, and power system.

Configured and troubleshot the electrical and flight-control systems to achieve successful flight.

Honors Awards

DECA Washington State Champion (2024, 2025) **FRC World Championship Qualifier (2023, 2024)** **FRC Sustainability Award (2024)**

Technical Skills

CAD & Mechanical: Onshape, SolidWorks, Mechanical Design, Robotics, Systems Integration

Manufacturing: CNC Machining, Manual Machining, Fabrication, Precision Machining

Programming: Python, C++, Java, Swift, JavaScript, React, HTML